

Microprocessor System Lab3 Check List

- Date: 2024/10/08
 - Please follow the instructions, and report to TAs after finishing check points.
 - If you have any questions or troubles during the experiment, please feel free to ask TAs or the teacher, discussion with classmates is also encouraged.
 - Please refer to the content from class slides, you can also use the resources from the internet.
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Part I. Question

Please refer to the class content and answer the following questions.

Check Points:

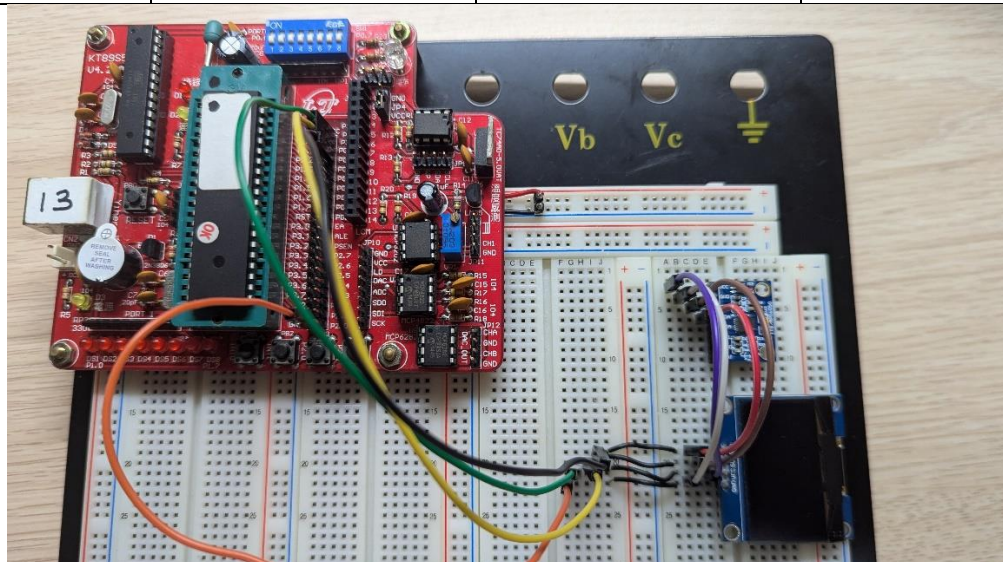
1.1: When OLED refresh the screen, it doesn't refresh the whole screen simultaneously while working as marquee. What's the reason behind?

Part II. Lab3 Experiment

In this experiment, we have two parts should be done, they are OLED and MPU6050 respectively. Source codes are already attached in the folder, you can try to write the code by yourself or use them directly to see the result.

Pins are as follows.

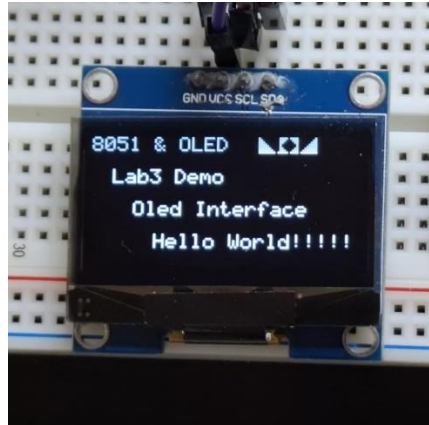
	燒錄器 pin	OLED pin	MPU6050 pin
Line 1	GND	GND	GND
Line 2	VCC	VCC	VCC
Line 3	P0.0	SCL	SCL
Line 4	P0.1	SDA	SDA



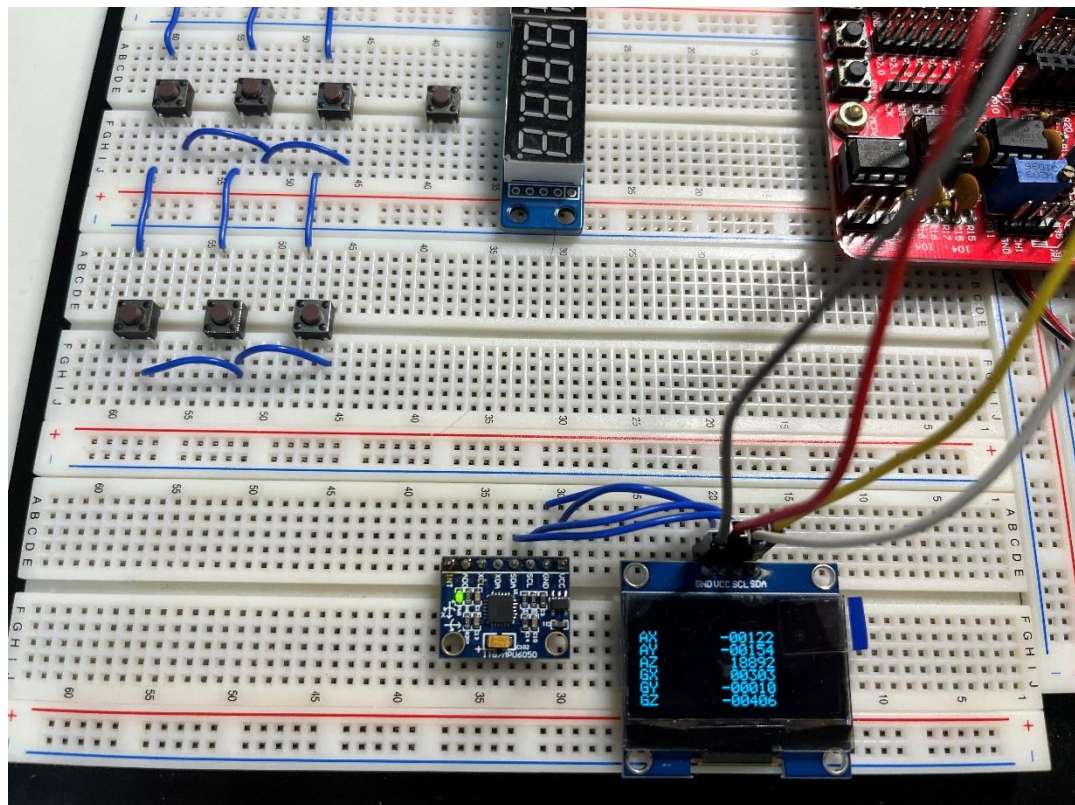
Check Points:

2.1: Connect wires between 8051 and OLED, execute "main.c" in the folder, "OLED". Observe what happened to the OLED and understand the code in Example project. You can refer to the lecture slides for detail execution.

Pins are as follows.



2.2: Connect the wires between 8051 ,MPU6050 and OLED module, execute the “main.c” in the folder, “OLED_MPU6050”. Observe what happened to the OLED and understand the code in “Example project”. You can refer to the lecture slides for detail execution.

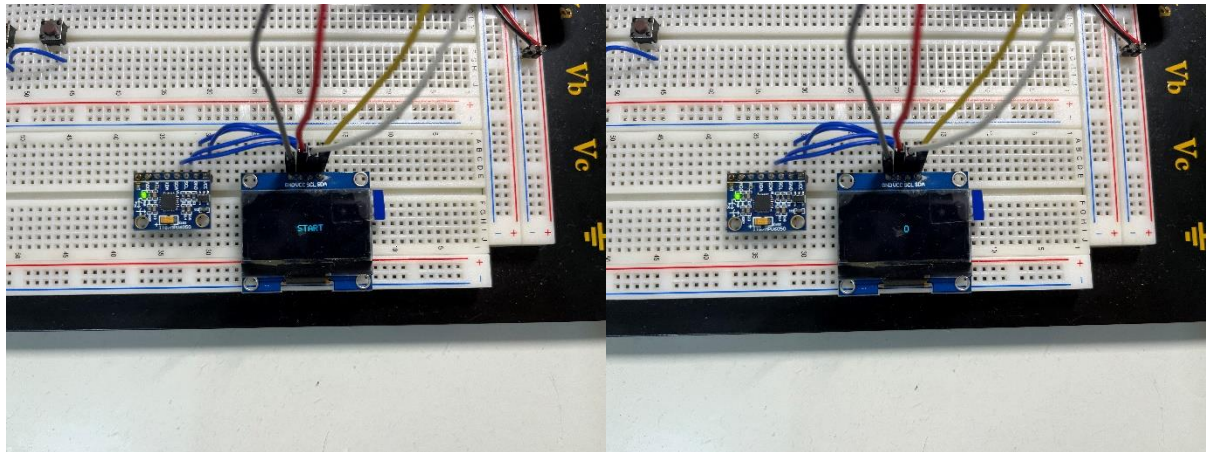


Part III. Practice

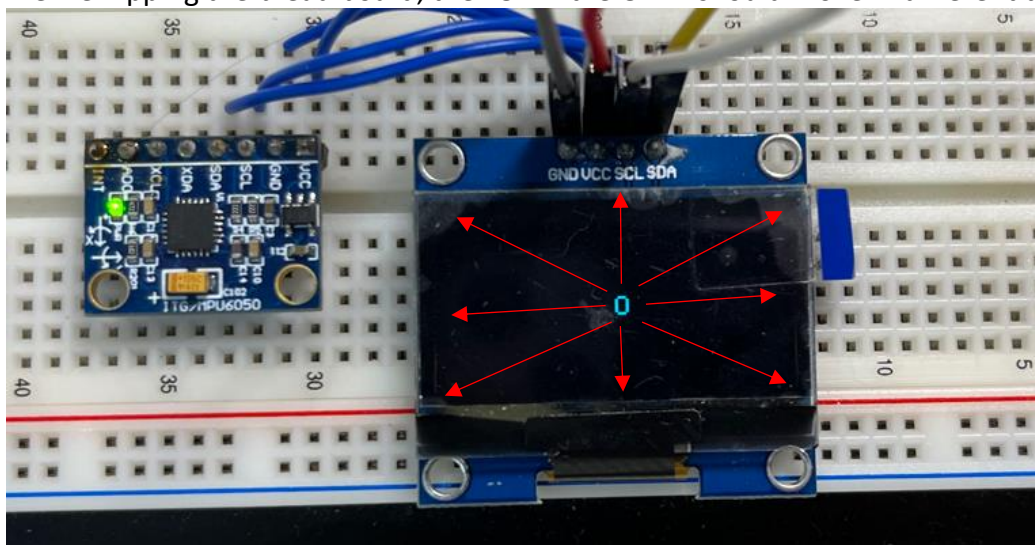
Please refer to the class content and the source code of Part II and complete the Practice. You can finish the practice using the code we’ve helped you started in the **Practice folder**, or start on your own, as long as it is well functioned.

Description:

3.1 Display “start” in the middle of the OLED screen for a while. And then display “O” in the center of the OLED screen, and let it shift as we flipping the MPU6050 module.



While we flipping the bread board, the “O” in the OLED should move in different direction



Check Points:

3.1: Function in Part III is correct.